

Testing VPC Connectivity

Melvin green

```
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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC, or Virtual Private Cloud, is a service that lets you create your own logically isolated network within AWS where you can deploy resources like EC2 instances and databases. It's

useful because it gives you full control over your network environment — you define your own IP address ranges using CIDR blocks, create public and private subnets, configure route tables, set up internet gateways, and control traffic using Security Groups and Network ACLs.

How I used Amazon VPC in this project

In this project, I used Amazon VPC to establish connectivity between my EC2 instances. I launched both a public and a private EC2 instance within the same VPC, configured dedicated security groups for each, and set up the networking so that only authorized SSH traffic could reach the private instance. This gave me hands-on experience with how VPC networking controls communication between resources in a cloud environment.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was that the ping command inside an EC2 instance runs continuously and doesn't stop on its own. In my previous experience outside of the cloud, ping typically sends around 4 packets and stops automatically. In AWS EC2, I had to manually cancel it using Ctrl+C. It was a small but important difference that reminded me that cloud environments don't always behave the same way as traditional on-premise systems

NextWork Student

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This project took me...

This project took me around an hour and a half.

Connecting to an EC2 Instance

Connectivity means Connectivity is all about how well different parts of your network communicate with each other and with external networks. Without connectivity, resources cannot communicate.

My first connectivity test was whether I could connect to my public server and EC2 instance.

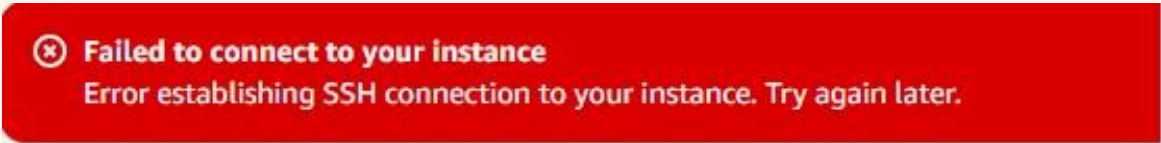


EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is a tool provided by Amazon EC2 that allows us to directly access an EC2 Instance using the AWS Management console! We no longer need to manage key pairs or use an SSH client to connect to our EC2 Instance.

My first attempt at getting direct access to my public server resulted in an error because my private server had a security group that did not allow SSH traffic, and it only allowed HTTP traffic, which has a different protocol.

I fixed this error by adding a new inbound rule in my private server security group that allows SSH traffic from anywhere.

A screenshot of an error message from the AWS Management console. The message is displayed in a red box with white text. It includes a circular icon with a white 'x' on a red background, followed by the text 'Failed to connect to your instance' and 'Error establishing SSH connection to your instance. Try again later.'

 **Failed to connect to your instance**
Error establishing SSH connection to your instance. Try again later.

Connectivity Between Servers

Ping is a tool to test the connectivity between two servers and also the response times. I used ping to test the connectivity between my public and private servers.

The ping command I ran was ping "10.0.1.0" where 10.0.1.0 is the private IPv4 address of my private server address for our private server.

The first ping returned NO replies from the private server. This meant the security setting with my private server was blocking inbound and or outbound ICMP traffic, which is the traffic type of ping messages.

[illegible]

Troubleshooting Connectivity

I troubleshooted this by enabling ICMP traffic in my private server's network ACLs and security group rules. As a bonus, I also made sure the source I defined in my network ACL correctly pointed to my public subnet.

```

~\##### Amazon Linux 2023
~\#####
~\#####|
~\#/
~V~'--> https://aws.amazon.com/linux/amazon-linux-2023
~
~
~/m/'
Last login: Mon Jun 8 00:56:30 2026 from 18.206.107.29
[ec2-user@ip-10-0-0-228 ~]$ ping 10.0.1.20
PING 10.0.1.20 (10.0.1.20) 56(84) bytes of data.
64 bytes from 10.0.1.20: icmp_seq=85 ttl=127 time=0.286 ms
64 bytes from 10.0.1.20: icmp_seq=86 ttl=127 time=0.267 ms
64 bytes from 10.0.1.20: icmp_seq=87 ttl=127 time=0.284 ms
64 bytes from 10.0.1.20: icmp_seq=88 ttl=127 time=0.278 ms
64 bytes from 10.0.1.20: icmp_seq=89 ttl=127 time=0.253 ms
64 bytes from 10.0.1.20: icmp_seq=90 ttl=127 time=0.287 ms
64 bytes from 10.0.1.20: icmp_seq=91 ttl=127 time=0.264 ms
64 bytes from 10.0.1.20: icmp_seq=92 ttl=127 time=0.273 ms
64 bytes from 10.0.1.20: icmp_seq=93 ttl=127 time=0.334 ms
64 bytes from 10.0.1.20: icmp_seq=94 ttl=127 time=3.37 ms
64 bytes from 10.0.1.20: icmp_seq=95 ttl=127 time=0.274 ms
64 bytes from 10.0.1.20: icmp_seq=96 ttl=127 time=0.290 ms
64 bytes from 10.0.1.20: icmp_seq=97 ttl=127 time=0.274 ms
64 bytes from 10.0.1.20: icmp_seq=98 ttl=127 time=0.271 ms
64 bytes from 10.0.1.20: icmp_seq=99 ttl=127 time=0.268 ms
64 bytes from 10.0.1.20: icmp_seq=100 ttl=127 time=0.286 ms
64 bytes from 10.0.1.20: icmp_seq=101 ttl=127 time=0.268 ms
64 bytes from 10.0.1.20: icmp_seq=102 ttl=127 time=0.276 ms
64 bytes from 10.0.1.20: icmp_seq=103 ttl=127 time=0.288 ms
64 bytes from 10.0.1.20: icmp_seq=104 ttl=127 time=0.298 ms
64 bytes from 10.0.1.20: icmp_seq=105 ttl=127 time=0.288 ms
64 bytes from 10.0.1.20: icmp_seq=106 ttl=127 time=0.272 ms
64 bytes from 10.0.1.20: icmp_seq=107 ttl=127 time=0.257 ms
64 bytes from 10.0.1.20: icmp_seq=108 ttl=127 time=0.260 ms
64 bytes from 10.0.1.20: icmp_seq=109 ttl=127 time=0.301 ms
64 bytes from 10.0.1.20: icmp_seq=110 ttl=127 time=0.271 ms
64 bytes from 10.0.1.20: icmp_seq=111 ttl=127 time=0.286 ms

```

Connectivity to the Internet

Curl is a connectivity tool that test connecivity from a server from a server to another server and retrieves data from the target server too.

I used curl to test the connectivity between my public server and the public internet. This test would only be successful if my internet gateway network ACLs, security groups, and route tables were set up correctly.

Ping vs Curl

Ping and curl are different because they return different responses to my public server's terminal - ping response with a report on the performance of the connectivity with my private servers, curl responded with HTML data from another public private server.

Connectivity to the Internet

I ran the curl command `curl https://learn.nextwork.org/projects/aws-host-awebsite-on-s3`, which returned the HTML content of NextWork's first project guide.

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